



ALLIANCE
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NAAC
GRADE A+
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 5

Title: Building Sales Performance Dashboard

Submitted By:

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Date: 5-8-2026

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Bengaluru – 562106

Experiment No.: 5

This preview may be up to 2 days old. Refresh

Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Order Date", type date}, {"Ship Date", type date}, {"Customer Name", type text}, {"Country", type text}, {"City", type text}, {"State", type text}})

	Order ID	Order Date	Ship Date	Customer Name	Country	City	State
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	CA-2013-138688	13-06-2013	17-06-2013	Darrin Van Huff	United States	Los Angeles	California
2	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
3	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
4	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
5	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
6	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
7	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
8	CA-2011-115812	09-06-2011	14-06-2011	Brosina Hoffman	United States	Los Angeles	California
9	CA-2013-161389	06-12-2013	11-12-2013	Irene Maddox	United States	Seattle	Washington
10	CA-2011-167164	13-05-2011	15-05-2011	Alejandro Grove	United States	West Jordan	Utah
11	CA-2011-143336	27-08-2011	01-09-2011	Zuschuss Donatelli	United States	San Francisco	California
12	CA-2011-143336	27-08-2011	01-09-2011	Zuschuss Donatelli	United States	San Francisco	California
13	CA-2011-143336	27-08-2011	01-09-2011	Zuschuss Donatelli	United States	San Francisco	California
14	CA-2012-106320	25-09-2012	30-09-2012	Emily Burns	United States	Orem	Utah
15	CA-2013-121755	16-01-2013	20-01-2013	Eric Hoffmann	United States	Los Angeles	California
16	CA-2013-121755	16-01-2013	20-01-2013	Eric Hoffmann	United States	Los Angeles	California
17	CA-2013-101343	18-07-2013	23-07-2013	Ruben Ausman	United States	Los Angeles	California
18	CA-2012-135545	24-11-2012	30-11-2012	Kunst Miller	United States	Los Angeles	California
19	CA-2012-135545	24-11-2012	30-11-2012	Kunst Miller	United States	Los Angeles	California
20	CA-2012-135545	24-11-2012	30-11-2012	Kunst Miller	United States	Los Angeles	California
21	CA-2012-135545	24-11-2012	30-11-2012	Kunst Miller	United States	Los Angeles	California
22	CA-2011-106376	05-12-2011	10-12-2011	Brendan Sweed	United States	Gilbert	Arizona
23	CA-2011-106376	05-12-2011	10-12-2011	Brendan Sweed	United States	Gilbert	Arizona

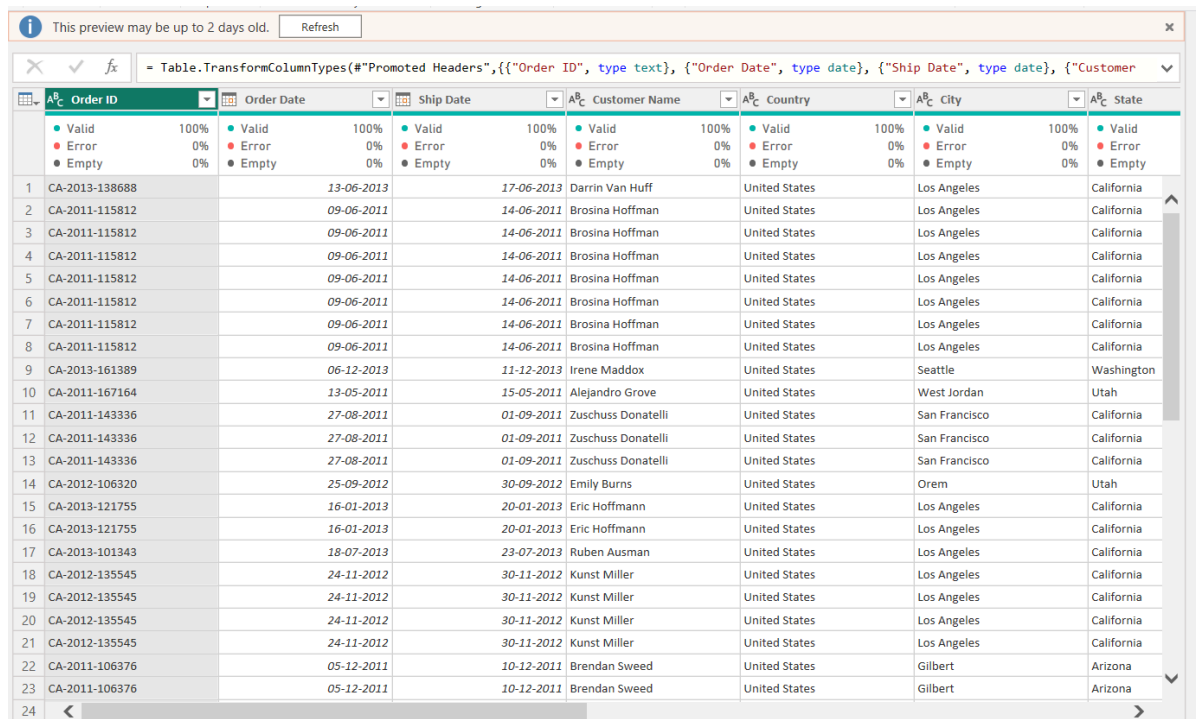
Step 2: Data Preparation

After importing the dataset, I verified the data types of all columns. The following columns were checked and converted into appropriate data types:

- Store → Whole Number
- Date → Date
- Weekly_Sales → Decimal Number
- Holiday_Flag → Whole Number
- Temperature → Decimal Number
- Fuel_Price → Decimal Number
- CPI → Decimal Number
- Unemployment → Decimal Number

No missing values or data errors were found during the preparation process.

Screenshot 2:



This preview may be up to 2 days old. Refresh

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Step 3: Data Preparation

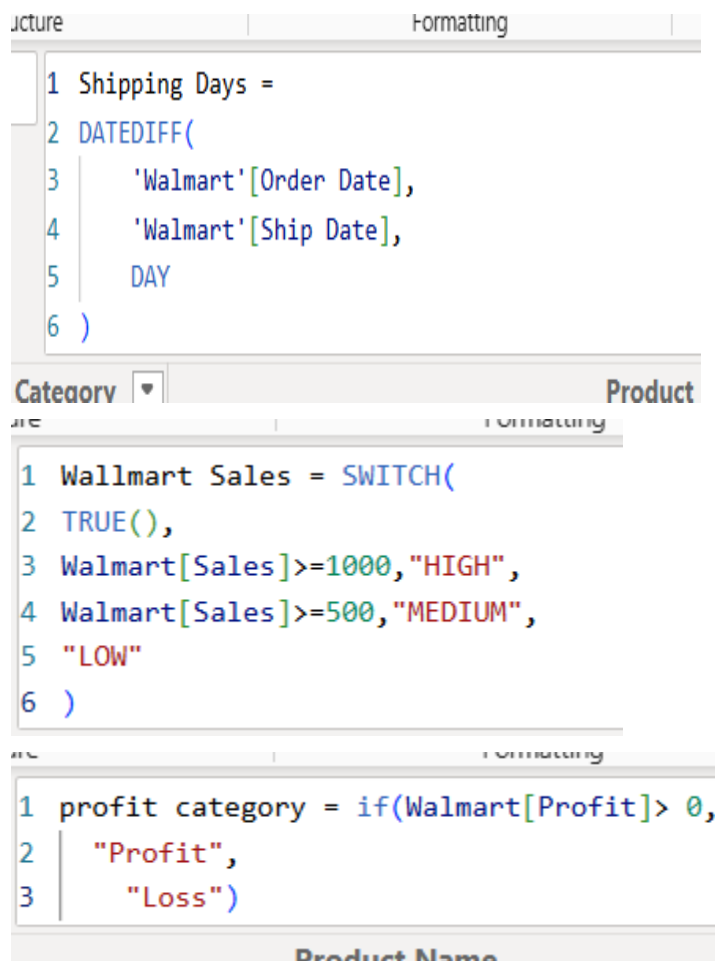
After importing the dataset, I verified the data types and prepared the data for analysis. I also created calculated columns to improve business analysis.

The following calculated columns were created:

- **Profit Category** – Classifies products based on profit.
- **Shipping Days** – Calculates the number of days between Order Date and Ship Date.
- **Walmart Sales** – Categorizes sales into Low, Medium, and High.

These columns help analyze product performance and sales more effectively.

Screenshot 3:



The image displays three screenshots of the Power BI DAX formula editor, showing the creation of calculated columns. The first screenshot shows the formula for 'Shipping Days', which uses the DATEDIFF function to calculate the difference in days between the 'Order Date' and 'Ship Date' columns for the 'Walmart' category. The second screenshot shows the formula for 'Walmart Sales', which uses the SWITCH function to categorize sales into 'HIGH', 'MEDIUM', or 'LOW' based on specific thresholds. The third screenshot shows the formula for 'profit category', which uses the IF function to categorize products as 'Profit' or 'Loss' based on the 'Profit' column values.

```
1 Shipping Days =  
2 DATEDIFF(  
3     'Walmart'[Order Date],  
4     'Walmart'[Ship Date],  
5     DAY  
6 )
```

```
1 Walmart Sales = SWITCH(  
2     TRUE(),  
3     Walmart[Sales]>=1000,"HIGH",  
4     Walmart[Sales]>=500,"MEDIUM",  
5     "LOW"  
6 )
```

```
1 profit category = if(Walmart[Profit]> 0,  
2     "Profit",  
3     "Loss")
```

Step 4: Create DAX Measures

To display important business metrics, I created the following DAX measures:

Total Sales

Total Sales = SUM(Walmart[Sales])

Total Profit

Total Profit = SUM(Walmart[Profit])

Total Quantity

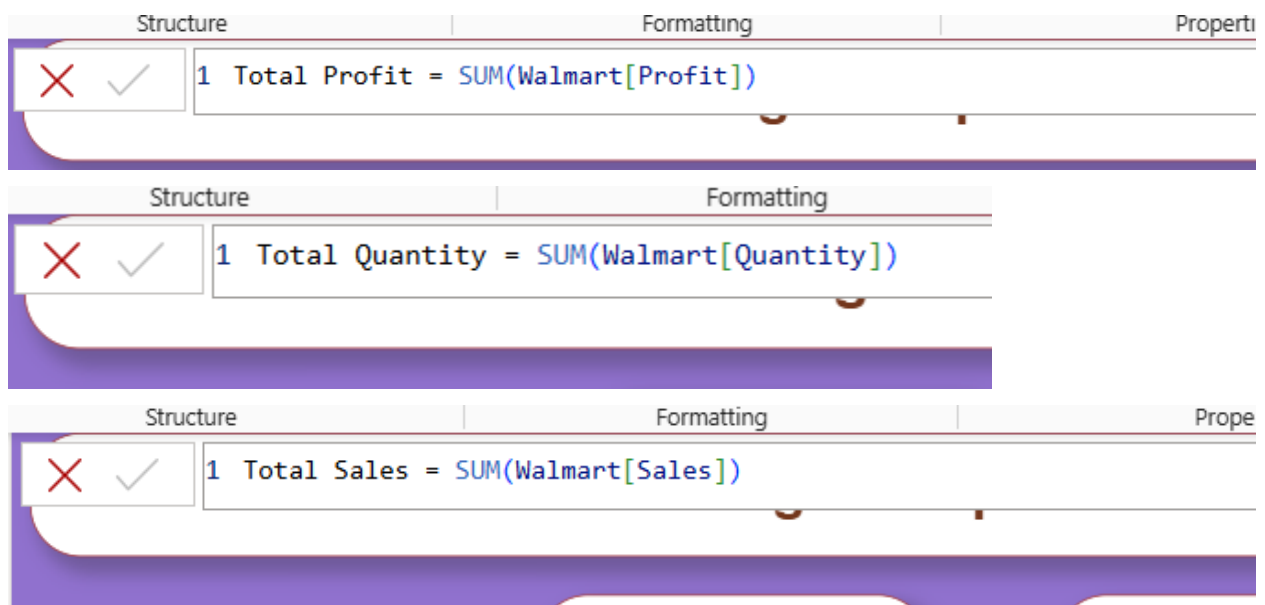
Total Quantity = SUM(Walmart[Quantity])

Total Orders

Total Orders = DISTINCTCOUNT(Walmart[Order ID])

These measures were used to create KPI cards on the dashboard.

Screenshot 4:



Steps 5: Dashboard Development

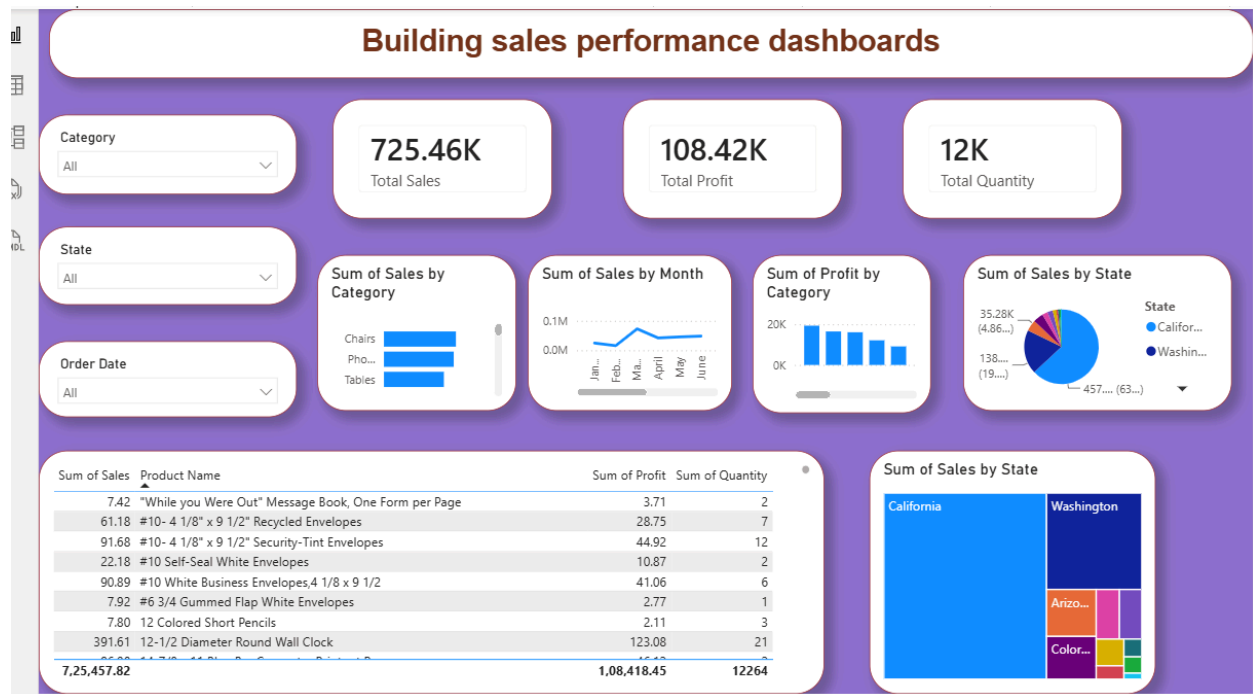
After preparing the dataset and creating measures, I designed an interactive sales dashboard in Power BI.

The dashboard contains the following visualizations:

- **KPI Cards** showing Total Sales, Total Profit, and Total Quantity.
- **Bar Chart** showing Sales by Category.
- **Line Chart** showing Monthly Sales Trend.
- **Column Chart** showing Profit by Category.
- **Pie Chart** showing Sales by State.
- **Treemap** showing State-wise Sales Distribution.
- **Table** displaying Product Name, Sales, Profit, and Quantity.
- **Slicers** for Category, State, and Order Date to filter the dashboard interactively.

These visualizations help management understand overall sales performance and business trends.

Screenshot 5:



Business Insights

From the dashboard, the following observations were made:

- The company generated 725.46K in total sales.
 - The total profit earned was 108.42K.
 - A total quantity of 12K products was sold.
 - Some product categories contributed more to sales than others.
 - Monthly sales showed fluctuations throughout the year.
 - California recorded the highest sales among all states.
 - The dashboard allows users to filter data by category, state, and order date for detailed analysis.
-

Conclusion

In this assignment, an interactive Sales Performance Dashboard was successfully developed using Power BI. The Walmart Sales dataset was imported, prepared, and analyzed using DAX measures and different visualizations. KPI cards and charts made it easy to monitor sales, profit, quantity, product performance, and regional sales. The dashboard provides meaningful business insights and helps management make informed decisions based on data.

Learning Outcomes

Through this assignment, I learned:

- How to import a business dataset into Power BI.
- How to prepare and organize data for analysis.
- How to create calculated columns and DAX measures.
- How to build KPI cards for business metrics.
- How to create Bar Charts, Line Charts, Column Charts, Pie Charts, and Treemaps.
- How to use slicers to make dashboards interactive.
- How to analyze sales performance and generate business insights using Power BI.

